THE INEQUALITY [GAMMA_B : GAMMA_A] TIMES [KAPPA_B : KAPPA_A] AT MOST [L : K] FOR A FINITE EXTENSION OF VALUATION RINGS

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Editorial note. The remark that precedes the lemma at source lines 37330-37331, recording that the extension of fraction fields is not required to be finite, is outside the counted spans because it belongs to the surrounding discussion and not to the lemma or its proof.

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Allowed standard prerequisites (editorial).

- A finite family of elements of a valued field with pairwise distinct valuations cannot sum to zero: after a minimal valuation is normalized to zero, the remaining terms have strictly larger valuation while a unit term cannot be cancelled by them. used at proof lines 37353-37371, where the index set P of minimal valuation is introduced, the coefficients are normalized and the sum over P is shown to be nonzero. Supplies the contradiction that proves the products b_i c_j are K-linearly independent.
- If b_1, ..., b_n are units whose residue classes in kappa_B are linearly independent over kappa_A and the coefficients a_ij are units, then the sum of the a_ij b_i over the index set P has valuation zero, hence is a unit of B. used at proof lines 37362-37366. Identifies the valuation of the partial sum over P with the valuation of the chosen minimal term, which is what makes the whole sum nonzero.
- Nonzero elements c_1, ..., c_m of B whose images in Gamma_B over Gamma_A are pairwise distinct force, for a fixed minimal valuation, the equality j = j_0 of the second index. used at proof lines 37346-37361.

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Reduces the index set P of minimal valuation to a single value of j, which is what allows the residue classes of the b_i to be used in the next step.

Lemma 0.2. *Let $A \subset B$ be an extension of valuation rings with fraction fields $K \subset L$. If the extension L/K is finite, then the residue field extension is finite, the index of Γ_A in Γ_B is finite, and*

$$[\Gamma_B : \Gamma_A][\kappa_B : \kappa_A] \leq [L : K].$$

Proof. Let $b_1, \ldots, b_n \in B$ be units whose images in κ_B are linearly independent over κ_A. Let $c_1, \ldots, c_m \in B$ be nonzero elements whose images in Γ_B/Γ_A are pairwise distinct. We claim that $b_i c_j$ are K-linearly independent in L. Namely, we claim a sum

$$\sum a_{ij} b_i c_j$$

with $a_{ij} \in K$ not all zero cannot be zero. Choose (i_0, j_0) with $v(a_{i_0 j_0} b_{i_0} c_{j_0})$ minimal. Replace a_{ij} by $a_{ij}/a_{i_0 j_0}$, so that $a_{i_0 j_0} = 1$. Let

$$P = \{(i, j) \mid v(a_{ij} b_i c_j) = v(a_{i_0 j_0} b_{i_0} c_{j_0})\}$$

By our choice of $c_1, \ldots, c_m$ we see that $(i, j) \in P$ implies $j = j_0$. Hence if $(i, j) \in P$, then $v(a_{ij}) = v(a_{i_0 j_0}) = 0$, i.e., a_{ij} is a unit. By our choice of $b_1, \ldots, b_n$ we see that

$$\sum\nolimits_{(i,j) \in P} a_{ij} b_i$$

is a unit in B. Thus the valuation of $\sum_{(i,j) \in P} a_{ij} b_i c_j$ is $v(c_{j_0}) = v(a_{i_0 j_0} b_{i_0} c_{j_0})$. Since the terms with $(i, j) \notin P$ in the first displayed sum have strictly bigger valuation, we conclude that this sum cannot be zero, thereby proving the lemma. $\square$